ANALYSTLAB AFRICA

Week 3 — FinTrust Digital Bank: Develop & Integrate

Field	Detail
Intern Name	Tatenda Farai Mujati
Track	Data Analytics
Week	Week 3
Data Source	Full FinTrust_Customer_Data.xlsx (1,500 records) + FinTrust_Transaction_Data.xlsx (12,000 records) — unchanged from Weeks 1–2

Part A — Review of Week 2 Analysis

Week 2 delivered a data-quality assessment, 8 SQL business questions, 5 Python visualisations, a working Power BI dashboard and 5 business findings. Reviewing that work against the full dataset surfaced three specific gaps worth addressing in Week 3 — one of which is a genuine methodological error, not just an enhancement opportunity.

Issue / Gap	Evidence	Required Improvement
Week 2's geographic finding conflated customer home city with transaction location	Week 2 joined transactions to the customer's City field and concluded 'Lagos accounts for 36.1% of value.' Re-running the analysis using the transaction's own Location field shows 87.3% of transactions occur in a location different from the customer's home city, and value is actually spread almost evenly across all 8 cities (11.1%–14.5% each) — Lagos is NOT the top location by this measure.	Re-validate the finding using the correct field (Location, not City) and revise the business recommendation accordingly — done in Part E, Finding 1 below.
Week 2's 'USSD is undervalued' finding was based on average value per transaction, not value at risk	Week 2 highlighted USSD's high average transaction value as a reliability-investment priority. Advanced SQL (AQ6) shows Web actually loses the highest % of its total value to Failed/Reversed transactions (8.24%), not USSD (7.32%).	Refine the channel-reliability recommendation to be based on value-at-risk, not average transaction size — done in Part E, Finding 2.
Week 2 dashboard lacked trend depth, customer-level analysis and interactivity beyond basic slicers	The Week 2 dashboard had 4 charts and 4 slicers but no cross-filtering validation, no customer-segment analysis, and only a daily (noisy) trend view.	Extend the dashboard with weekly trend analysis, customer segmentation views, risk-by-segment analysis and validated cross-filtering between visuals — delivered in Part D below.

Part B — Advanced SQL Analysis (Summary)

Full queries, results and interpretations are in FinTrust_Week3_Advanced_SQL_Analysis.sql — 8 new queries using window functions (RANK, NTILE, LAG, SUM OVER), CTEs, subqueries and CASE logic, extending beyond Week 2's single-table aggregations. Headline new findings:

- The top 10% of transactions by value account for 56.15% of all transaction value (AQ4, NTILE decile analysis) — the single strongest new finding this week.
- Premium and Student customer segments both sit above the average risk-review rate (20.33% and 19.75% vs. 19.60% overall) — not visible in Week 2's channel-only risk analysis (AQ3).

- Monthly trend analysis (AQ2) shows a real February dip (−₦12.7m vs January) followed by a March recovery (+₦20.4m) — Week 2's daily view looked flat and masked this.
- 73.13% of customers have at least one transaction exceeding 3x their own personal average (AQ7) — a validation check showing per-customer transaction-size spikes are too common to be a useful standalone risk signal.
- International transaction risk rate exceeds domestic in every single month (AQ8: 41.6%, 36.2%, 33.2% vs. 17.6%, 19.7%, 19.5%), confirming the Week 2 finding is stable — though the gap is visibly narrowing.

Part C — Advanced Python Analysis (Summary)

Full notebook: FinTrust_Week3_Advanced_Analysis.ipynb — 5 new analyses extending Week 2's EDA: weekly trend analysis, a Pareto/value-concentration chart, customer segmentation (engagement vs. frequency), a comparative international-vs-domestic risk trend, and a correlation validation check.

- The Pareto curve visually confirms the SQL AQ4 finding: a steep concentration where 10% of transactions carry 56% of value.
- Digital Engagement Score and transaction frequency are essentially uncorrelated (r = −0.0 to 0.02 across checks) — a validated negative finding meaning engagement campaigns should not assume higher engagement automatically drives higher transaction volume.
- Age, Tenure_Months and Digital_Engagement_Score are mutually independent (correlation heatmap, all values near zero) — useful confirmation that these are non-redundant features for any future modelling work.

Part D — Power BI Dashboard (Extended)

The Week 2 dashboard was extended with three new sections — Trend Analysis, Customer Analysis and an expanded Risk Review Analysis, built on the same cleaned data export. Power BI's default cross-filtering between visuals was explicitly tested and confirmed (see Testing & Interactivity Evidence below) rather than assumed to work. An interactive version mirroring the intended Power BI structure is also published for cross-checking: it runs the same KPI logic, charts and filters live against all 12,000 transactions.

What was added beyond Week 2

- Weekly transaction value trend (replacing/supplementing the noisier daily view) — surfaces the February dip found in Part B.
- Customer Analysis section: average engagement score by segment, and risk-review rate by segment — neither existed in the Week 2 dashboard.
- Cross-filtering between visuals: clicking a bar on any chart (e.g. a channel on Risk Rate by Channel) filters the KPI cards and every other visual on the page to match the selection — Power BI's built-in interactivity, explicitly tested rather than assumed (see Testing & Interactivity Evidence below).
- Risk Review Analysis expanded from one chart (by channel) to two (by channel and by segment), directly reflecting the Part B finding that risk varies meaningfully by segment, not just by channel.

Testing & Interactivity Evidence

Dashboard interactivity was explicitly tested rather than assumed to work, per the Week 3 testing requirement:

Test	Finding	Action	Result
Clicked a channel on Risk Rate by Channel	KPI cards and other visuals responded to the selected channel	Confirmed default visual interactions	Dashboard cross-filtering works correctly

Part E — Validate Business Findings

Finding 1: Lagos geographic concentration

Original finding (Week 2): Lagos accounts for 32.6% of transaction volume and 36.1% of total value, a significant geographic concentration worth prioritising for infrastructure investment.

Evidence re-examined: Re-querying using the transaction's own Location field (not the customer's home City) shows value is spread almost evenly across all 8 cities (11.1%–14.5% each), with Ibadan actually highest by this measure (14.5%) — not Lagos. Separately, 87.3% of transactions occur in a location different from the customer's home city.

Validation: Does NOT support the original finding as stated. The Week 2 conclusion was correct about customer residence (36.1% of FinTrust's customers by value do live in Lagos) but incorrect as a statement about where transactions are processed — the two are different questions that happened to get conflated.

Revised business meaning: Lagos is where a disproportionate share of valuable customers live, useful for marketing, acquisition and relationship-management targeting. It is NOT where transaction processing load concentrates — that is spread evenly nationally, so infrastructure/reliability investment should not be Lagos-first on these grounds.

Finding 2: USSD channel value priority

Original finding (Week 2): USSD carries the highest average transaction value and should not be deprioritised for reliability investment.

Evidence re-examined: Advanced SQL (AQ6) measuring % of channel value lost to Failed/Reversed status shows Web at 8.24% value-at-risk, the highest of any channel — USSD sits lower at 7.32%.

Validation: Partially supported. USSD's high average value is still accurate and real, but it is not the channel losing the largest share of its value to failures — Web is.

Revised business meaning: two distinct priorities exist and should not be merged into one recommendation: USSD merits attention because of what it's worth per transaction; Web merits attention because of how much of its value is actually being lost.

Finding 3: International transactions carry disproportionate risk (VALIDATED)

Original finding (Week 2): international transactions are flagged for risk review at roughly double the domestic rate.

Evidence re-examined: Month-by-month breakdown (SQL AQ8, Python Viz 4) confirms international risk rate exceeds domestic in every one of the three months (41.6% / 36.2% / 33.2% vs. 17.6% / 19.7% / 19.5%).

Validation: Fully supported, and now shown to be stable across time rather than a one-off aggregate figure.

Business meaning (updated): the finding holds, but the gap is narrowing over the quarter — worth monitoring into Week 4, since a continued narrowing trend would eventually change how much relative priority international transactions deserve.

Finding 4: Transaction value is highly concentrated (NEW, validates an implicit Week 2 assumption)

Original finding: Week 2 treated all transactions as roughly equal priority for operational attention.

Evidence: SQL AQ4 and the Python Pareto chart both show the top 10% of transactions by value account for 56.15% of all transaction value.

Validation: This reveals an implicit gap in Week 2's framing rather than correcting a stated finding — Week 2 never tested whether transactions should be weighted equally.

Business meaning: reliability and risk-review resources produce more impact per unit of effort when weighted toward the highest-value transactions, not spread evenly across all transaction counts.

Finding 5: Top-value customers are not necessarily 'Premium' (VALIDATED & EXTENDED)

Original finding (Week 2): 3 of the top 5 customers by transaction value are labelled 'Everyday', not 'Premium'.

Evidence re-examined: SQL AQ1 (ranking customers within their own segment) confirms this pattern holds structurally, not just among the overall top 5 — each segment has its own internal high-value customers who may not be recognised as such under FinTrust's current segment labelling.

Validation: Fully supported and extended — this is not a quirk of 5 specific customers, it is a systematic pattern visible within every segment.

Business meaning: a per-segment, rank-based view (as built in AQ1) is a more actionable tool for relationship management than a single global top-5 list, since it surfaces high performers FinTrust might otherwise overlook within 'lower-tier' segments.

Part F — Management Recommendations

Finding	Evidence	Business Implication	Recommended Action
Geographic concentration was mis-attributed to Lagos	87.3% of transactions occur outside the customer's home city; Location-based value is evenly spread (11–14.5% per city)	Infrastructure investment decisions based on the Week 2 Lagos finding would have been misdirected	Base infrastructure/reliability investment decisions on the transaction Location field, not customer home City; reserve City-based analysis for marketing/acquisition questions only

Finding	Evidence	Business Implication	Recommended Action
Web, not USSD, has the highest value-at-risk	8.24% of Web's transaction value is Failed/Reversed, vs 7.32% for USSD	Reliability engineering effort aimed at USSD alone would miss the channel losing the most actual value	Prioritise Web reliability investigation first, with USSD retained as a secondary priority given its high average transaction value
International transactions carry consistently elevated risk, but the gap is narrowing	Risk rate gap: 41.6% → 33.2% (international) vs 17.6% → 19.5% (domestic) over 3 months	The current 'international = top priority' risk rule may need recalibrating if the trend continues	Keep international transactions as the top risk-review priority for now, but re-run this specific trend check in Week 4 before finalising any permanent risk-weighting rule
56% of transaction value sits in the top 10% of transactions	NTILE decile analysis (SQL AQ4), confirmed visually via Pareto chart	Equal-weighted operational attention under-serves the transactions that matter most financially	Introduce value-weighted monitoring thresholds (e.g. closer review for transactions above a value percentile) rather than treating all transactions as equal priority
High-value customers exist within every segment, not just 'Premium'	Within-segment ranking (SQL AQ1) shows top performers in Everyday, Student and SME segments alongside Premium	Current segment labels may cause relationship teams to overlook high-value customers outside the 'Premium' label	Build a per-segment top-customer report (as demonstrated in AQ1) into regular relationship-management reporting, rather than relying on segment label alone to identify high-value customers

Validation & Refinement Requirement

The chain below documents the most significant evidenced improvement made this week, following the required structure: Initial Output → Test → Finding → Refinement → Re-test → Result.

Stage	Detail
Initial Output	Week 2 business finding: “Lagos accounts for 32.6% of transaction volume and 36.1% of total value — a significant geographic concentration,” based on joining transactions to the customer's City field.

Stage	Detail
Test	Re-ran the geographic aggregation in Week 3 using the transaction's own Location field instead of the customer's City field, and separately checked how often the two fields actually agree.
Finding	The two fields disagree 87.3% of the time. Using Location (the methodologically correct field for 'where did this transaction occur'), transaction value is spread almost evenly across all 8 cities (11.1%–14.5% each) — Lagos is not even the top city by this measure.
Refinement	Revised Finding 1 in Part E to separate two distinct questions that Week 2 had conflated: "where do valuable customers live" (City — Lagos-concentrated, still valid) versus "where is transaction value processed" (Location — evenly spread, not Lagos-concentrated). Updated the Part F recommendation accordingly, removing the unsupported Lagos-first infrastructure suggestion.
Re-test	Cross-checked the Location-based city breakdown against the raw transaction count by Location to confirm the even spread wasn't an artefact of a calculation error — counts (1,431–1,550 per city) confirmed the same even pattern as the value totals.
Result	A materially more accurate geographic finding and recommendation, and a documented methodological lesson (always confirm which of two similarly-named fields a business question actually requires) carried forward into the Week 4 dashboard refinement plan.

Assumptions, Limitations, Risks & Decisions — Week 3 Update

Type	Item	Status
Resolved from Week 2	Geographic finding based on City vs Location was unverified	Resolved — tested, found to be a real conflation, and corrected in Part E/F (see Validation & Refinement section above)
Resolved from Week 2	Channel reliability priority (USSD vs Web) was based on average value only	Resolved — refined using value-at-risk, see Finding 2
New limitation	Dashboard's City-based geography chart is now known to measure customer residence, not transaction location	Documented as a Week 4 item (add a Location-based view) rather than rebuilt mid-week to avoid rushed errors
New observation	International risk-rate gap vs domestic is narrowing month over month	Flagged as something to re-check with Week 4 data before finalising any permanent risk-weighting policy

Type	Item	Status
Carried forward from Week 2	192 cells (1.6%) of transaction rows have missing Device_Type/Location, filled with 'Unknown'	Still applies; not affected by this week's work since the City/Location finding concerns correctness of a field already present for all rows, not missingness
Carried forward from Week 2	Account_Type='Premium' vs Customer_Segment='Premium' naming collision	Still applies; no further action needed beyond the explicit field-qualification already adopted in all Week 3 queries

Week 3 Project Summary

1. Planned: review Week 2 for gaps, extend SQL and Python analysis with more advanced techniques, expand the Power BI dashboard and validate 5 Week 2 findings with supporting evidence.

2. Completed: 8 advanced SQL queries (window functions, CTEs, subqueries), 5 additional Python analyses, an extended dashboard with two new analysis sections and tested cross-filtering, 5 validated findings (one fully revised, one refined, three confirmed) and 5 evidence-based recommendations.

3. Major development activity: re-examining the Week 2 geographic finding using the correct underlying field (Location vs City) — this was the most significant single piece of work this week, not just an incremental addition.

4. Key findings: top 10% of transactions carry 56% of total value; Lagos concentration was a City/Location conflation and does not hold for transaction processing location; Web (not USSD) has the highest value-at-risk from failed/reversed transactions; international risk rate exceeds domestic every month but the gap is narrowing; high-value customers exist within every segment, not just Premium.

5. Testing/validation performed: all 5 Week 2 findings were re-tested against fresh queries rather than assumed correct; the Lagos finding specifically was cross-checked twice (by value and by transaction count) before being revised.

6. Improvements made: corrected the geographic finding and its downstream recommendation; refined the channel-reliability priority from a single metric (average value) to a more decision-relevant one (value-at-risk); expanded the dashboard from 4 to 6 charts and explicitly tested cross-filtering interactivity.

7. Major challenge: discovering that a Week 2 finding was methodologically flawed, not just incomplete — required deciding whether to quietly fix it or document the correction transparently; chose transparency, since the reasoning behind a correction is itself evidence of analytical rigor.

8. Important decision: did not rebuild the dashboard's City-based chart mid-week after discovering the Location discrepancy, to avoid introducing a rushed error close to the deadline — documented as a deliberate, planned Week 4 item instead.

9. Remaining limitations: the dashboard still needs a Location-based geographic view; the narrowing international-risk-gap trend needs re-confirmation with a full additional month of data in Week 4.

10. What must be completed in Week 4: add the Location-based geographic dashboard view; re-check the international risk trend; finalise documentation and prepare for presentation/integration.

11. Final-week priorities: focus Week 4 on validation depth and presentation clarity rather than new analysis breadth — the core analytical work is now complete and evidenced; Week 4 is about confirming it holds up and communicating it clearly.